Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 38: Approved Solutions

Adjacency Matrices and Families of Graphs

Covers Chartrand & Zhang, Section 12.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Three checks on any adjacency matrix

1. The diagonal is all 0.
2. It is symmetric.
3. Row i sums to $\deg v_i$, and all entries sum to $2m$.

If any check fails, you have made an error — or it is not an adjacency matrix.

Powers count walks

The (i, j) entry of A^k is the number of v_i–v_j walks of length k. In particular the diagonal of A^2 is the degree sequence.

Three mistakes to expect

1. **Confusing walks with paths.** A^k counts walks, which may repeat.
2. **Forgetting symmetry** when filling the matrix in.
3. **Reading the size as the entry sum.** That sum is $2m$.

Part II: Practice Problems

Section A — Building and reading

1. Let $V = \{v_1, v_2, v_3, v_4\}$ and $E = \{v_1v_2, v_1v_3, v_1v_4, v_2v_3\}$.

(a) Write the adjacency matrix A.

(b) Give the degree sequence and the size, reading both from A.

Approved solution

(a)

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(b) Row sums are $3, 2, 2, 1$, so that is the degree sequence. The entries total 8, which is $2m$, so $m = 4$.
✓ (Diagonal all zero, matrix symmetric — both checks pass.)

2. For the matrix A above, compute A^2 and use it to answer:

(a) How many walks of length 2 join v_2 and v_4?

(b) Why is the $(1, 1)$ entry of A^2 equal to 3?

Approved solution

$$A^2 = \begin{bmatrix} 3 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

(a) The $(2, 4)$ entry is 1: the single walk (v_2, v_1, v_4).
(b) The $(1, 1)$ entry counts closed walks of length 2 at v_1 — go out along an edge and come straight back. There is one per edge at v_1, so the entry is $\deg v_1 = 3$.

The $(1, 4)$ entry is 0: no walk of length 2 joins v_1 and v_4, since they are adjacent and have no common neighbour. **Adjacent does not imply a length-2 walk.**

Section B — Families

3. Give the order and size of each.

(a) K_8

(b) $K_{4,5}$

(c) C_9

(d) P_9

Approved solution

(a) Order 8, size $\binom{8}{2} = 28$. **(b)** Order 9, size $4 \cdot 5 = 20$.
(c) Order 9, size 9. **(d)** Order 9, size 8.

Note (c) and (d): a cycle has as many edges as vertices, a path has one fewer. Deleting any one edge of C_9 gives P_9.

4.

(a) How many graphs are there on 5 labelled vertices?

(b) Is C_6 bipartite? Is C_7? Justify without finding a split.

Approved solution

(a) A graph is a subset of the $\binom{5}{2} = 10$ possible edges, so there are $2^{10} = 1024$.
(b) C_6 **is** bipartite and C_7 is **not**. A graph is bipartite exactly when it has no odd cycle; C_6 is a single even cycle, while C_7 is itself an odd cycle.

Part (a) is Week 4 and Week 13 together: the edge set is a subset (power set, 2^n) of the pairs (a combination, $\binom{5}{2}$).

Section C — Writing

5. In four to six complete sentences, explain why the adjacency matrix of a graph is always symmetric with a zero diagonal, and what each would mean if violated.

Approved solution

Model answer. An edge is a two-element subset $\{u, v\}$, so saying u is adjacent to v is the same statement as saying v is adjacent to u; the matrix records the same fact at entry (i, j) and entry (j, i), which forces symmetry. A non-symmetric matrix would be recording a direction on each edge, which is a *directed* graph rather than a graph. The diagonal is zero because an edge has two *distinct* endpoints, so no vertex is adjacent to itself. A nonzero diagonal entry would mean a loop at that vertex, which the definition of a graph excludes but which a pseudograph permits.

Full credit requires naming what each violation would represent — a digraph and a loop. An answer that only restates the properties earns about half.

Instructor notes

- **Problem 2(b)** — diagonal of A^2 is the degree sequence — is the most useful fact here and the one students find surprising.
- **Problem 2's closing remark:** v_1 and v_4 are adjacent yet have no length-2 walk between them. Adjacency and short walks are different things.
- **Problem 4(a)** ties power sets to combinations. Worth naming both.
- **Problem 4(b)** rewards using the odd-cycle criterion rather than hunting for a split.
- Reserved for Homework 14: Section 12.1 even-numbered exercises.